Firm Characteristics and the Cross-Sectional Variation in Post-Earnings Announcement Drift

Colin DiPasquale

2026

Abstract

This paper explores which characteristics of a given firm have predictive power over the magnitude and persistence of post-earnings announcement drift (PEAD) using both a machine learning framework and linear regression approach. Using Compustat quarterly fundamentals and CRSP daily returns for a sample of 6,728 U.S. firms over 2004 to 2023, we construct a panel of 203,764 firm-quarter earnings events and compute cumulative abnormal returns (CARs) over 21 and 63 trading-day post-announcement windows. Standardized unexpected earnings (SUE) are measured via a seasonal random walk model. We then train a LightGBM gradient boosted model alongside an OLS ridge regression in a walk-forward validation framework, predicting 1 month post-announcement drift from 13 firm characteristics. The gradient boosted model achieves an out of sample R^2 of 1.04%, generating a predicted Q5-Q1 portfolio spread of 4.29% over 21 trading days, exceeding the raw SUE-sorted spread of 3.27%. Short term pre-announcement momentum, book-to-market ratio (BTM ratio), and intermediate momentum are the strongest predictors of drift magnitude, while SUE itself ranks ninth, suggesting that how a firm arrives at an earnings surprise matters more than how large the surprise is. The LGBM model (interchangeably referred to as the gradient boosted or machine learning model) substantially outperforms the linear

baseline (interchangeably referred to as the OLS model or ridge regression model, R^2 of -0.0003), indicating that the predictive relationships are fundamentally nonlinear.

Introduction

Post-earnings announcement drift is a well documented phenomenon in empirical finance. First discovered by Ball and Brown (1968) and formally characterized by Bernard and Thomas (1989, 1990), PEAD refers to the tendency of stock prices to continue drifting in the direction of an earnings surprise for weeks to months following an announcement, in apparent violation of the semi-strong regime of the efficient market hypothesis.

Surprisingly, despite decades of documentation and understanding, PEAD has not been fully arbitraged away. The literature attributes its persistence to limits of arbitrage; transaction costs, short-sale constraints, and investor inattention. These, among others, prevent institutional traders from fully correcting the mispricing. However, the cross-sectional variation in drift magnitude across firms remains underexplored. While it is well established that drift is stronger for smaller, less covered firms, the relative importance of a broader set of characteristics (and their nonlinear interactions) has not been systematically determined using modern machine learning methods.

This paper addresses that gap. The central research question is as follows: which firm characteristics, observable at the time of an earnings announcement, best predict the subsequent magnitude of post-announcement drift? We approach this question using gradient boosted trees (LightGBM), which can capture nonlinear relationships and feature interactions that standard OLS regression cannot. We evaluate both models using walk-forward out-of-sample validation to ensure no lookahead bias.

There are three main findings. First, PEAD is blatantly present in the sample: sorting on SUE produces a Q5-Q1 CAR spread of 3.27% over 21 trading days. Second, firm characteristics predict meaningful cross-sectional variation in drift beyond SUE alone: the LightGBM model generates a 4.29% Q5-Q1 spread over the same time period. Third, pre-announcement price momentum and firm valuation ratios lead the feature importance rankings, while SUE itself is only the ninth most important predictor.

Data and Sample Construction

Data Sources

The analysis draws from three primary datasets, all available through the Wharton Research Data Services (WRDS) platform. Quarterly earnings data and firm fundamentals are obtained from Compustat North America (comp.fundq), covering years 2000-2023. Daily stock returns are obtained from the CRSP Daily Stock File (crsp.dsf), restricted to common shares (CRSP share codes 10 and 11). Daily Fama-French three-factor returns are obtained from the Ken French data library, also from WRDS.

Earnings Surprise Measure

Standardized Unexpected Earnings (SUE) are computed using a seasonal random walk model, which requires no analyst forecast data. The raw earnings surprise for firm i in quarter t is defined as the change in basic earnings per share (epspxq) relative to the same fiscal quarter from the previous year. This surprise is then standardized by the firm's rolling 8 quarter standard deviation of seasonal EPS changes, with a floor of $0.01 to prevent division by zero or near-zero denominators. The resulting SUE is winsorized at the 1st and 99th percentiles. Winsorization was chosen to eliminate large outliers that could potentially skew data, though follow-up work may be done to consider outcomes without winsorization.

Sample

Starting from approximately 795,000 raw firm-quarter observations in Compustat, we apply standard filters: industrial format (INDL), standardized data (STD), domestic firms, consolidated statements, non-missing earnings announcement dates (rdq), and positive total assets and price. After requiring at least four prior quarters of history to compute valid SUE, the Compustat panel contains 462,716 firm-quarter observations. Matching to CRSP via CUSIP linkage yields a final event panel of 203,764 firm-quarter events across 6,728 unique firms, spanning 2004 to 2023. CUSIP linkage is simply a table used to connect firm IDs across datasets such as Compustat and CRSP. It does not provide any additional information.

Abnormal Returns

Cumulative abnormal returns (CARs) are computed using a market-adjusted model, where the daily abnormal return is the firm's raw return minus the CRSP value-weighted market return. Three event windows are constructed relative to the earnings announcement date: the announcement window (days -1 to 0), the short drift window (days +1 to +21, approximately one month), and the medium drift window (days +1 to +63, approximately three months). All CARs are winsorized at the 1st and 99th percentiles. Once again, further work without winsorization may be done to explore any meaningful effects.

PEAD Baseline Results

Table 1 presents mean CARs by SUE quintile for the full sample. The results confirm the presence of PEAD: firms in the highest SUE quintile (Q5, largest positive surprise) earn a mean 21-day CAR of +2.58%, while firms in the lowest quintile (Q1) earn -0.69%, yielding a Q5-Q1 spread of 3.27%. The pattern monotonically increases across quintiles and persists and expands over the 63-day window (3.92% spread). The positive Q5-Q1 spread confirms the strong presence of PEAD.

Table 1: Mean CARs by SUE Quintile

SUE Quintile	**Description**	**Mean CAR(+1,+21)**	**Mean CAR(+1,+63)**
Q1 (Most Negative)	Large earnings miss	-0.69%	+1.06%
Q2	Below expectations	-0.16%	+1.41%
Q3	In-line	+0.85%	+2.69%
Q4	Above expectations	+1.41%	+3.61%
Q5 (Most Positive)	Large earnings beat	+2.58%	+4.98%
Q5 – Q1 Spread		**+3.27%**	**+3.92%**

Note: CAR(+1,+21) and CAR(+1,+63) are market-adjusted cumulative abnormal returns over 21 and 63 trading days after the announcement. Sample: 203,764 firm-quarter events, 2004–2023.

Machine Learning Framework

Features

Thirteen firm characteristics are constructed, all observable at the time of the earnings announcement. These span five categories: (1) earnings surprise (SUE, absolute SUE); (2) size and liquidity (log market capitalization, log total assets, exchange listing); (3) valuation (book-to-market ratio, earnings-to-price ratio, financial leverage); (4) earnings quality (Sloan accruals, earnings volatility, consecutive beat streak); and (5) pre-announcement price momentum (6-month momentum skipping the most recent month, 1-month momentum). More work may be done with additional firm characteristics, though additional characteristics are likely to simply be derived from the thirteen aforementioned or are not common enough to create substantial datasets.

Models

Two models are created: Ridge Regression (OLS with L2 regularization, $\alpha = 1.0$) as a linear baseline, and LightGBM gradient boosted trees as the primary model. LightGBM is well-suited to this setting: it handles missing values natively, captures nonlinear feature interactions, is relatively simple to implement, and is computationally efficient on large tabular datasets. The entire project, available on GitHub, took roughly one hour to compute on an Intel i7-1260P with 32 gigabytes of memory.

Walk-Forward Validation

To avoid lookahead bias, models are evaluated using an expanding walk-forward framework. In each loop, the model is trained on all data prior to a given year and tested on that year's observations. With a minimum training window of five years, this yields 15 test loops covering 2009 to 2023 and 147,601 out-of-sample observations. Features are imputed (via median) and scaled using statistics computed on training data only.

Results

Model Performance

Table 2 presents out-of-sample model performance. LightGBM achieves an R^2 of .0104, consistent with the range typically reported in the return predictability literature (Gu, Kelly, and Xiu 2020 report similar magnitudes for monthly return prediction). Ridge regression achieves an R^2 of essentially zero (-0.0003), confirming that the predictive relationships are completely nonlinear. The LightGBM Q5-Q1 predicted portfolio spread of 4.29% exceeds both the OLS spread (3.07%) and the raw SUE baseline (3.27%), indicating that the model adds genuinely meaningful predictive insight beyond the earnings surprise itself.

Table 2: Out-of-Sample Model Performance

Model	Out-of-Sample R^2	RMSE	Q5–Q1 Spread
Ridge Regression (OLS)	−0.0003	0.1360	3.07%
LightGBM (Gradient Boosted Trees)	0.0104	0.1352	4.29%

Note: Walk-forward validation, 15 test folds (2009–2023), 147,601 out-of-sample observations. Q5-Q1 spread is the difference in mean actual CAR(+1,+21) between the highest and lowest predicted quintiles.

Portfolio Sort

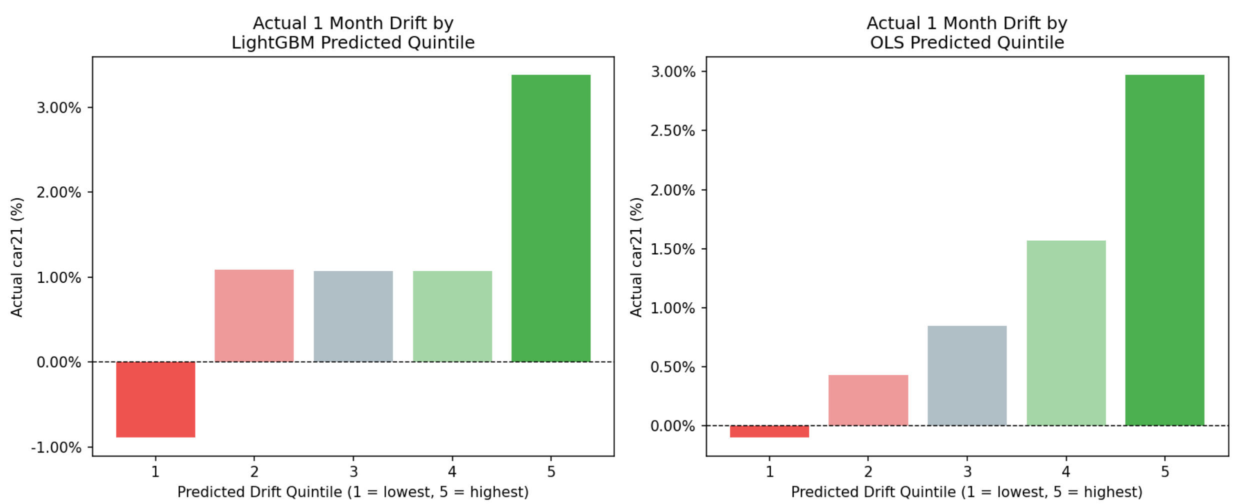

Figure 1: Actual 1-month CAR by predicted drift quintile. Left: LightGBM. Right: Ridge OLS. The LightGBM Q5−Q1 spread of 4.29% exceeds the OLS spread of 3.07%.

Figure 1 illustrates the portfolio sort results. The LightGBM sort (left panel) exhibits a notably nonlinear pattern: Q1 through Q4 are relatively compressed (ranging from −0.94% to +1.17%), while Q5 jumps sharply to +3.35%. This suggests that the model identifies distinct subsets of both high-drift and low-drift events characterized by a specific combination of features that a linear model cannot separate. The OLS sort (right panel) shows a more gradual monotonic pattern, with Q2 through Q4 spanning 0.43% and 1.57%. Winsorization may have diminished the results seen, hence why more work is suggested. However, if winsorization did have an effect, it would only have diminished the trend seen in the current data. In other words, winsorization can only have downplayed the nonlinearity of the relationship, and removing winsorization may increase the predictive power of the LightGBM model while decreasing the predictive power of the OLS model.

Feature Importances

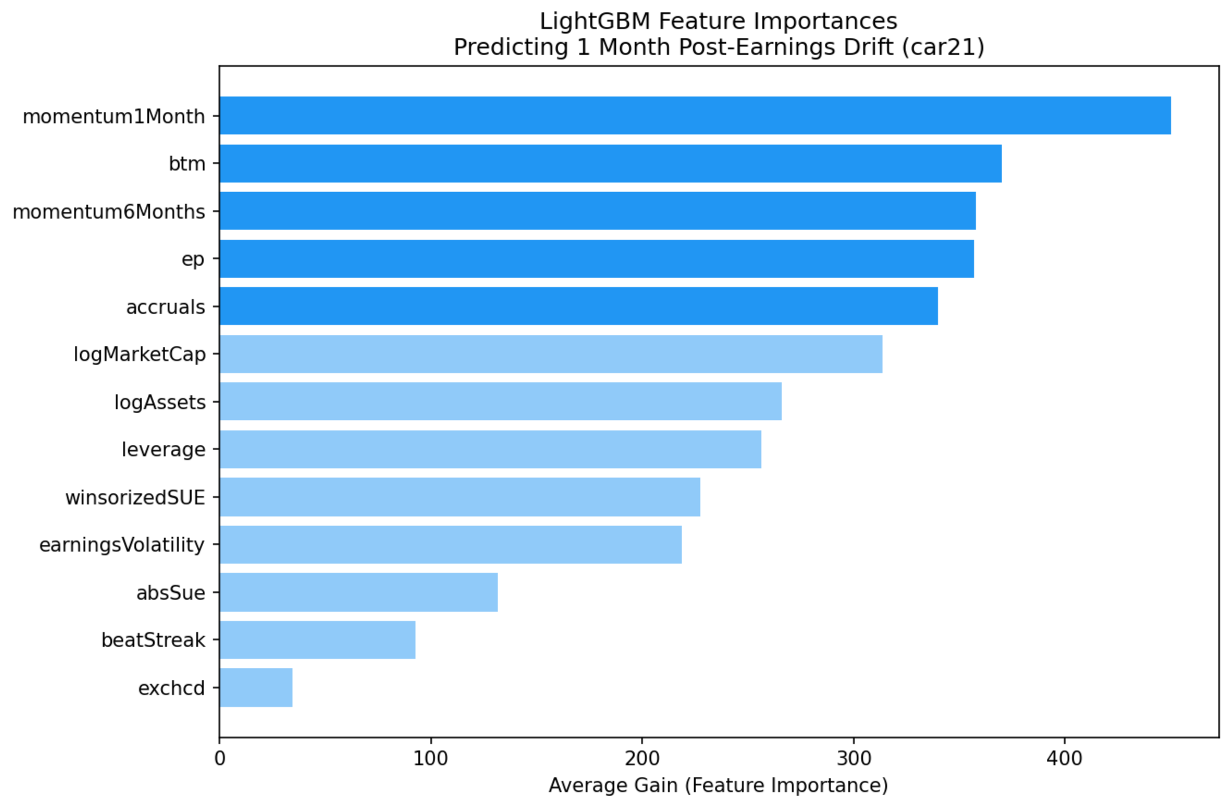

Figure 2: LightGBM feature importances (average gain across 15 walk-forward folds). Dark blue bars indicate the top-5 predictors.

Figure 2 presents feature importances measured by average gain (the average reduction in prediction error attributable to each feature across all splits). The results reveal a clear hierarchy. Short-term pre-announcement momentum (mom_m1, return in the month before announcement) is the strongest predictor, followed by book-to-market ratio (btm), intermediate momentum (mom_m6), earnings-to-price ratio (ep), and Sloan accruals. The earnings surprise measure itself (sue_wins) ranks ninth, with roughly half the importance of the top features. Exchange listing (exchcd) is nearly irrelevant after controlling for size. As mentioned in a previous section, future work may include more features, though there is a reasonable concern additional features will not add new insights as many are derived from existing features or are too infrequent to build a comprehensive model.

The dominance of momentum variables is consistent with the investor attention literature: stocks already attracting attention and trending prior to an announcement continue to exhibit stronger post-announcement drift, potentially reflecting informed trading or attention-driven underreaction. The high importance of valuation ratios (btm, ep) suggests that value firm characteristics interact meaningfully with PEAD, possibly through limits-to-arbitrage channels; value stocks tend to be smaller and less liquid, making drift harder to arbitrage away. The largest surprise was how low SUE ranked; this reveals that there is very little correlation between the size of a surprise and the post-announcement drift. However, this may not be true for the immediate (<<1 day) effects of a stock price, as this was not the focus of the paper. More work should be done to study the driving characteristics behind extremely short-term post-earning announcement effects.

Temporal Stability

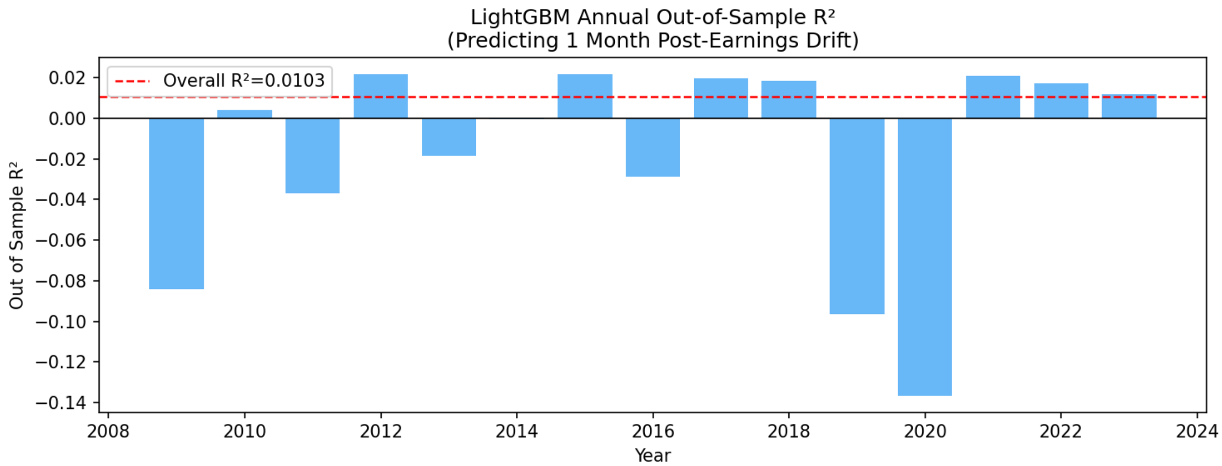

Figure 3: Annual out-of-sample R^2 (LightGBM). Red dashed line indicates the full-period R^2 of 0.0104. Large negative values in 2009 and 2020 correspond to post-crisis regime breaks.

Figure 3 shows annual R^2 over the test period. The model performs well in most years but breaks down sharply in 2009 and 2020 — the immediate aftermath of the Global Financial Crisis and the COVID-19 market dislocation, respectively. These are regime-change environments where historical cross-sectional relationships collapse. The model shows improving and consistent predictive performance from 2017 onward (excluding 2020), suggesting the relationships have stabilized in the post-crisis period. More work could be considered to evaluate R^2 without those years. While interesting to see, it was not done here as it would not accurately reflect the true nature of the model.

Conclusion

This paper documents that firm characteristics predict meaningful cross-sectional variation in post-earnings announcement drift beyond what is captured by the surprise alone. Using gradient boosted trees in a walk-forward validation framework, we show that pre-announcement price momentum, book-to-market ratio, and earnings quality measures are the strongest predictors of drift magnitude, while SUE ranks relatively low in feature importance (ninth overall). The nonlinear model generates a Q5-Q1 predicted portfolio spread of 4.29% over 21 trading days, exceeding the raw SUE-sorted spread of 3.27%.

These findings have implications for both asset pricing theory and quantitative practice. The dominance of momentum variables over the surprise measure itself suggests that the mechanism underlying PEAD is more closely related to investor attention and information diffusion than to the earnings signal. The failure of the linear model to capture any predictive content reinforces that the relevant relationships are nearly completely (if not completely) nonlinear.

Several limitations warrant acknowledgment. First, the CUSIP-based CRSP-Compustat link achieves a 55% match rate, potentially introducing survivorship bias toward larger, more established firms. Second, reported CARs are gross of transaction costs; for smaller firms where drift is strongest, bid-ask spreads and market impact costs may substantially reduce or eliminate net returns. Third, statistical significance of the Q5-Q1 spread is not formally tested using Newey-West standard errors; future work may be done here. Aside from winsorization, additional characteristics, and further error calculation, future work may also incorporate analyst forecast data (IBES) for a refined SUE measure, short interest data for arbitrage cost proxies, and SHAP values for more granular feature interaction analysis. A solution to the CRSP-Compustat link issue that yields a higher match rate can also be considered as future work, potentially eliminating any biases towards larger firms.

References

Ball, R., and P. Brown. 1968. An empirical evaluation of accounting income numbers. *Journal of Accounting Research* 6: 159–178.

Bernard, V., and J. Thomas. 1989. Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research* 27: 1–36.

Bernard, V., and J. Thomas. 1990. Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13: 305–340.

Gu, S., B. Kelly, and D. Xiu. 2020. Empirical asset pricing via machine learning. *Review of Financial Studies* 33: 2223–2273.

Ke, G., Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T. Liu. 2017. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems* 30.

Sloan, R. 1996. Do stock prices fully reflect information in accruals and cash flows about future earnings? *Accounting Review* 71: 289–315.